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Engineering the surface structure of MoS_2 to preferentially expose active edge sites for electrocatalysis

Jakob Kibsgaard†, Zhebo Chen†, Benjamin N. Reinecke and Thomas F. Jaramillo*

Controlling surface structure at the atomic scale is paramount to developing effective catalysts. For example, the edge sites of MoS_2 are highly catalytically active and are thus preferred at the catalyst surface over MoS_2 basal planes, which are inert. However, thermodynamics favours the presence of the basal plane, limiting the number of active sites at the surface. Herein, we engineer the surface structure of MoS_2 to preferentially expose edge sites to effect improved catalysis by successfully synthesizing contiguous large-area thin films of a highly ordered double-gyroid MoS_2 bicontinuous network with nanoscaled pores. The high surface curvature of this catalyst mesostructure exposes a large fraction of edge sites, which, along with its high surface area, leads to excellent activity for electrocatalytic hydrogen evolution. This work elucidates how morphological control of materials at the nanoscale can significantly impact the surface structure at the atomic scale, enabling new opportunities for enhancing surface properties for catalysis and other important technological applications.

Chalcogenide derivatives of molybdenum (Mo) have proved to be exceptionally promising and versatile building blocks for the design of nanomaterials in recent decades. Several stable one- or two-dimensional molybdenum sulphide nanostructures have been discovered with unique combinations of optical[1–3], mechanical[4,5], electronic[6] and chemical properties[7,8] completely different from those observed for the three-dimensional bulk molybdenite phase. Traditionally the interest in MoS_2 nanostructures stems from the environmentally and technologically important hydrodesulphurization (HDS) process where supported MoS_2 nanoclusters are employed to remove sulphur from crude oil feedstocks[9]. However, the unique properties of MoS_2 nanostructures have also promoted their use in fields as diverse as nanotribology[10], photocatalysis[11,12], solar cells[13,14] and recently hydrogen evolution reaction (HER) catalysis[15–18]. In a similar fashion to its activity for HDS catalysis, the HER activity for MoS_2 is correlated with the number of exposed edge sites that have a local stoichiometry, physical structure and electronic structure that differs from the catalytically inert (0001) basal planes of MoS_2 (ref. 15). Hence, a prerequisite for an active MoS_2-based HER catalyst is the ability to nanostructure MoS_2 to maximize the number of exposed edge sites relative to bulk sites. Previous studies have focused on: MoS_2 nanoparticles supported on Au (ref. 15), graphene[16], graphite[19], carbon paper[20], CdS (ref. 21) and TiO_2(ref. 22) or MoS_2 films supported on SiO_2(ref. 23), and doped SnO_2 (ref. 17). To improve the activity per geometric area, high-aspect-ratio architectures can be employed such as core–shell MoO_3–MoS_2 nanowires[18]. In the last case, transmission electron microscopy (TEM) images revealed that most of the MoS_2 basal planes were parallel to the nanowire axis, resulting in few edge sites exposed at the surface of the nanowires, limiting the HER activity on this high-surface-area structure. It is likely that a thermodynamic driving force impedes the formation of highly energetic and atomically undercoordinated edge sites at the surface relative to non-active basal sites. Consequently, few edge sites are found on bulk MoS_2 crystals, which consist of large extended sheets; similarly, nanoscaled forms of MoS_2 typically form closed-shell fullerene structures with few exposed edge sites.

To produce a more active catalyst, one needs to control the atomic-scale surface structure to preferentially expose a greater fraction of the sites that are catalytically active. This is particularly challenging when the formation of catalytically active edge sites can be thermodynamically unfavourable compared with the formation of inert basal plane sites[24]. Herein, we describe our efforts to develop a MoS_2 catalyst with a higher ratio of edge sites relative to basal plane sites by controlling the nanoscale morphology of the catalyst material. We accomplish this by synthesizing mesoporous MoS_2 thin films with a high degree of surface curvature. A MoS_2 morphology with a radius of curvature of nanoscale dimensions is expected to engender more exposed edge sites by imposing geometric constraints that can markedly change the thermodynamic stabilities of different surface features[24]. Here, on a particular mesoporous MoS_2 morphology, we show that the formation of extended basal plane structures is hindered to a large extent in favour of catalytically active edge sites, leading to improved catalytic activity.

Another important requirement in the design of a mesoporous catalyst is sufficient transport of reactants and products, for example, H^+ and H_2 in the case of HER, through the high-surface-area thin-film structure (that is, minimal diffusion limitations). To maximize the diffusivity, we synthesize an extended three-dimensional mesoporous MoS_2 network structure with double-gyroid morphology, illustrated in Fig. 1. We selected this structure as previous studies on platinum and SiO_2 have shown that the double-gyroid mesostructure exhibits a diffusivity that is higher by an order of magnitude compared with other mesoporous structures[25]. We report the first synthesis of a fully contiguous large-area thin film of mesoporous MoS_2. In contrast to other

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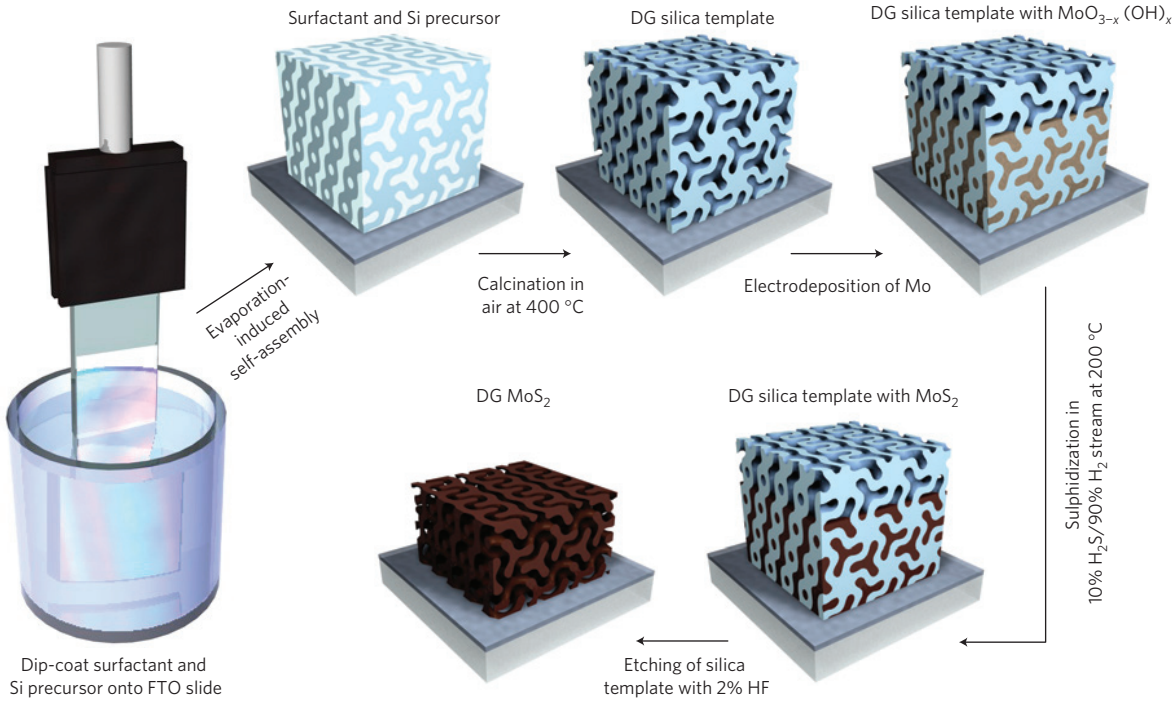

Figure 1 | Synthesis procedure and structural model for mesoporous MoS_2 with a double-gyroid (DG) morphology.

morphologies of mesoporous materials (powders, fibres, monoliths and so on)[26–29], supported thin films offer the distinct advantage of ready integration into a device architecture for electronic, optical, sensing and, in the present case, electrochemical applications[30,31].

The double-gyroid MoS_2 structure was synthesized by electrodeposition of Mo (see below for details) into a silica template using various deposition times to control the film thickness, followed by sulphidization with H_2S. The silica double-gyroid template was prepared using EO_{19}–PO_{43}–EO_{19} surfactant (Pluronic P84, BASF) following the procedure developed in ref. 32. Tetraethyl orthosilicate (TEOS) was prehydrolysed by mixing with an aqueous solution of HCl and ethanol. The EO_{19}–PO_{43}–EO_{19} surfactant in ethanol was added to the mixture to form the coating solution. This solution was dip-coated onto fluorine-doped tin oxide (FTO, sheet resistance: $8\,\Omega\,\square^{-1}$, Hartford Glass) substrates and dried at a controlled relative humidity of ~50%, which enabled evaporation-induced self-assembly of the surfactant into the double-gyroid structure.

After evaporation was complete, the films were calcined in air at 400 °C for 4 h. The calcined silica templates with thicknesses of approximately 300 nm as measured by profilometry (Supplementary Fig. S1) were analysed using low-angle X-ray diffraction (XRD) and TEM to ensure that they exhibited the double-gyroid structure. Figure 2a shows the small-angle XRD spectrum of the silica template. The XRD pattern for the calcined double-gyroid silica templates exhibits four Bragg diffraction peaks. The films are highly oriented with their (211) planes parallel to the substrate as indicated by the main peak located at $2\theta = 1.93°$ (ref. 32). Three smaller peaks at lower 2θ values are also present with increasing 2θ values arising from the $(12\bar{1})$, $(2\bar{1}1)$ and (121) planes. These smaller peaks are important fingerprint patterns used to distinguish the double-gyroid structure from hexagonal or lamellar phases[32]. TEM images of the silica template further confirmed the double-gyroid morphology. Figure 2b shows a TEM image of the silica double-gyroid template. The left part of Fig. 2b exhibits the hexagonal pattern indicative of the [111] projection whereas the wavy striped pattern to the right is indicative of the [211] projection.

Molybdenum was electrodeposited from a peroxo-polymolybdate solution into the silica template using a standard

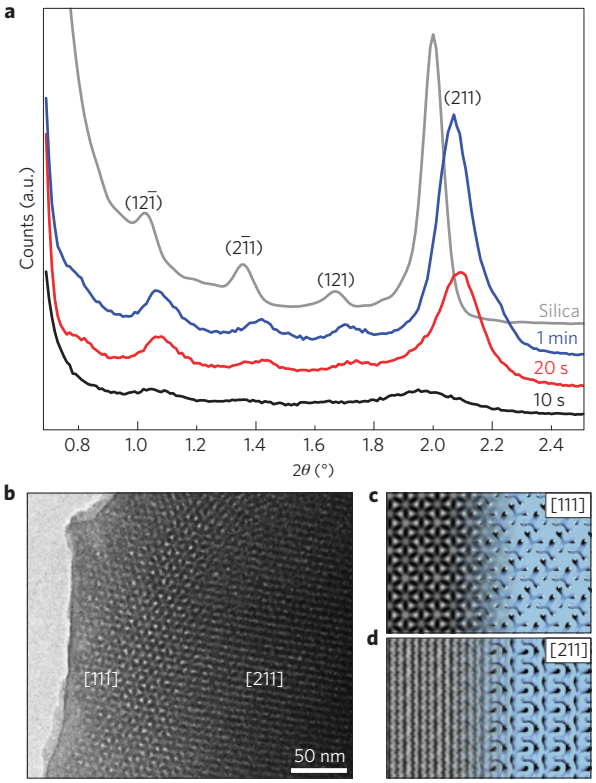

Figure 2 | Structural characterization of the double-gyroid morphology. **a**, Low-angle XRD of the double-gyroid silica template (intensity scaled × 1/2) and double-gyroid MoS_2 with various Mo electrodeposition times. All diffractograms exhibit the four fingerprint Bragg diffraction peaks for the double-gyroid structure. **b**, TEM image of the silica double-gyroid template. The hexagonal pattern in the left portion is indicative of the [111] projection whereas the wavy striped pattern in the right portion is indicative of the [211] projection. **c**,**d**, Models and corresponding simulated TEM images of the [111] and [211] projections, respectively.

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three-electrode electrochemical set-up (plots of passed charge versus time are presented in Supplementary Fig. S2). The chemical nature of the deposited Mo was investigated by X-ray photoelectron spectroscopy (XPS; Supplementary Fig. S3a). Deconvolution of the XPS spectra by peak fitting revealed three Mo 3*d* doublets indicative of a mixed-valent molybdenum bronze ($Mo^{VI/V/IV}O_{3-x}(OH)_x$), in agreement with previous studies of cathodic electrodeposition of molybdenum oxide thin films[33]. However, this molybdenum bronze is not stable in HF and etching of the silica template at this step leads to complete dissolution of the film. Sulphidization of the molybdenum bronze while still encapsulated in the silica template using a 10%/90% H_2S/H_2 gas mixture at 200 °C converts the film into a MoS_2 composition capable of withstanding the HF etching process[27].

The chemical states of Mo and S were investigated by XPS (Supplementary Fig. S3b,c) after etching away the silica template from the films. Most of the Mo signal arises from a $3d_{5/2}$ peak at 229 eV characteristic of a +4 oxidation state; shoulders corresponding to higher +5 and +6 oxidation states remain, probably due to the formation of some surface oxide after HF treatment and exposure to air. This is further confirmed by the presence of a shoulder in the S 2*p* region at higher binding energies. The samples were exposed to a second round of sulphidization at 200 °C to convert residual oxide into sulphide. From the XPS spectra in Fig. 3 and Supplementary Fig. S3d,e, the +4 oxidation state of Mo is clearly dominant for all samples after this second round of sulphidization, whereas the S 2*p* region exhibits primarily a single doublet with the $2p_{3/2}$ peak at 162 eV consistent with a −2 oxidation state for the sulphur. A two-step sulphidization therefore enables the surface to be composed primarily of MoS_2 as measured by XPS. Raman spectroscopy further confirmed the MoS_2 structure with the appearance of two distinct peaks at 380 cm^{-1} and 405 cm^{-1} for the E^1_{2g} and A_{1g} vibrational modes, respectively[34,35] (Fig. 3c).

The small-angle XRD data of the mesoporous MoS_2 thin films are shown in Fig. 2a. The electrodeposited Mo film nucleates at the FTO surface, where it can form electrical contact, and as it grows upwards it fills the three-dimensional bicontinuous network pore structure of the non-conductive silica double-gyroid template. Thus, the electrodeposited film represents the negative of the silica morphology, as illustrated in Fig. 1. Accordingly, the peak positions match those of the template, confirming the double-gyroid nature of the MoS_2 in accordance with Babinet's principle[36]. We observe a small shift to larger 2θ values for double-gyroid MoS_2 relative to the peak positions of the double-gyroid silica template, indicating that the double-gyroid MoS_2 structure is slightly contracted relative to the double-gyroid silica template, possibly owing to a slight difference in the size of the negative morphology or contraction on conversion of the molybdenum bronze into MoS_2. Another factor is the age of the template, which we observed to shift towards larger 2θ values with time (Supplementary Fig. S4). The intensities of the diffraction peaks of the double-gyroid MoS_2 are smaller relative to the silica template because the electrodeposited films underfilled the silica template (that is, less material to diffract X-rays). Peak intensities for the double-gyroid MoS_2 increase with longer electrodeposition times, as expected owing to the formation of a thicker film.

The thin-film nature of the double-gyroid MoS_2 was further investigated by scanning electron microscopy (SEM). Figure 4 shows SEM images of the MoS_2 double-gyroid structure. The mesoporous MoS_2 structure is stable after removal of the silica and retains the interlocked, bicontinuous twisted network structure, forming a thin uniform conformal coating on the FTO support. The grainy structure of the FTO is clearly visible under the thin layer of double-gyroid MoS_2 for the sample synthesized using a 10 s deposition time. As the deposition time increases, the film

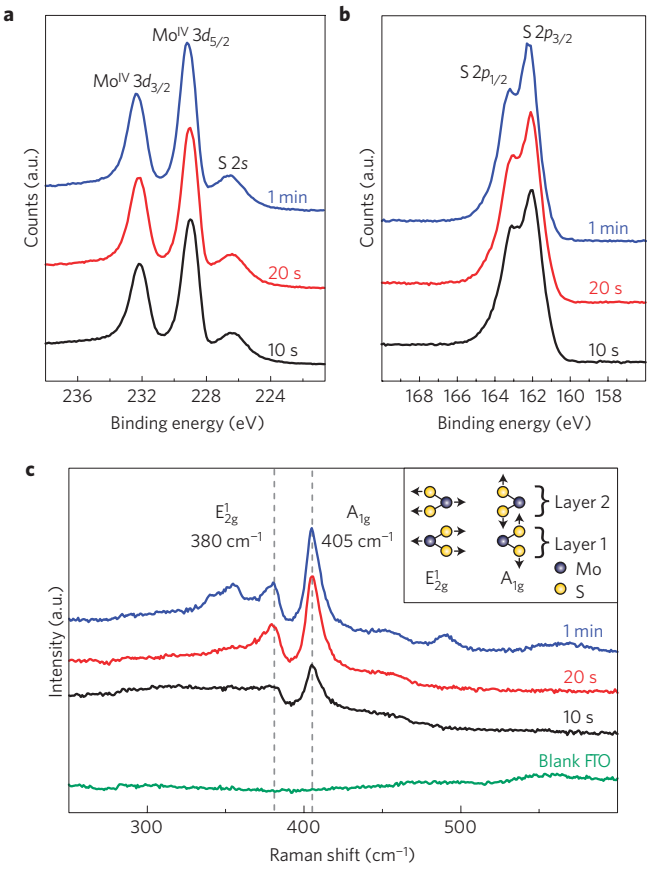

Figure 3 | Spectroscopic characterization of double-gyroid MoS_2 from various Mo electrodeposition times. a,**b**, High-resolution XPS spectra of the Mo 3*d* and S 2*s* peaks (**a**) and S 2*p* peaks (**b**). **c** ,Raman spectra showing the E^1_{2g} and A_{1g} vibrational modes. The inset shows the atomic displacements of the E^1_{2g} and A_{1g} vibrational modes in bulk MoS_2 as viewed along the [1000] direction.

gradually becomes smoother and the underlying FTO structure disappears. The cross-sectional images in Fig. 4e–h reveal that the film thicknesses range from tens to hundreds of nanometres. Thicker films also seem darker by eye, as evidenced by ultraviolet–visible transmission measurements (Supplementary Fig. S5). The TEM images in Fig. 5 magnify the mesoporous double-gyroid MoS_2 structure and illustrate the high porosity of these thin films. The double-gyroid nature of the MoS_2 is clearly evidenced in Fig. 5a where the [311] and [211] projections of the double-gyroid structure are observed. A closer analysis of the [211] structure in Fig. 5a,d shows that the distance between the MoS_2 networks (and pore-to-pore distance) is 7 nm and the thickness of the individual MoS_2 network is ∼4 nm (∼3 nm wide pores). Figure 5f shows a TEM image where the S–Mo–S layers are resolved and the inset linescan shows that the layer-to-layer spacing is 6.6 Å as indicated in the model. The bulk structure of MoS_2 consists of S–Mo–S trilayers separated by a Van der Waals gap and a layer-to-layer spacing of 6.15 Å (ref. 37). We attribute this slightly larger layer-to-layer spacing relative to bulk MoS_2 as a consequence of being confined to the double-gyroid morphology, which results in curvature of the MoS_2. This effect was previously observed for MoS_2 nanowires[18]. XRD, SEM and TEM images thus confirm that we have highly porous, mesostructured MoS_2 with a double-gyroid morphology.

We evaluated the electrochemical activity of double-gyroid MoS_2 for the HER. The HER activity data (per projected geometric area) of double-gyroid MoS_2 with Mo deposition times of 10 s, 20 s and 1 min are shown in Fig. 6a as cyclic voltammograms taken at

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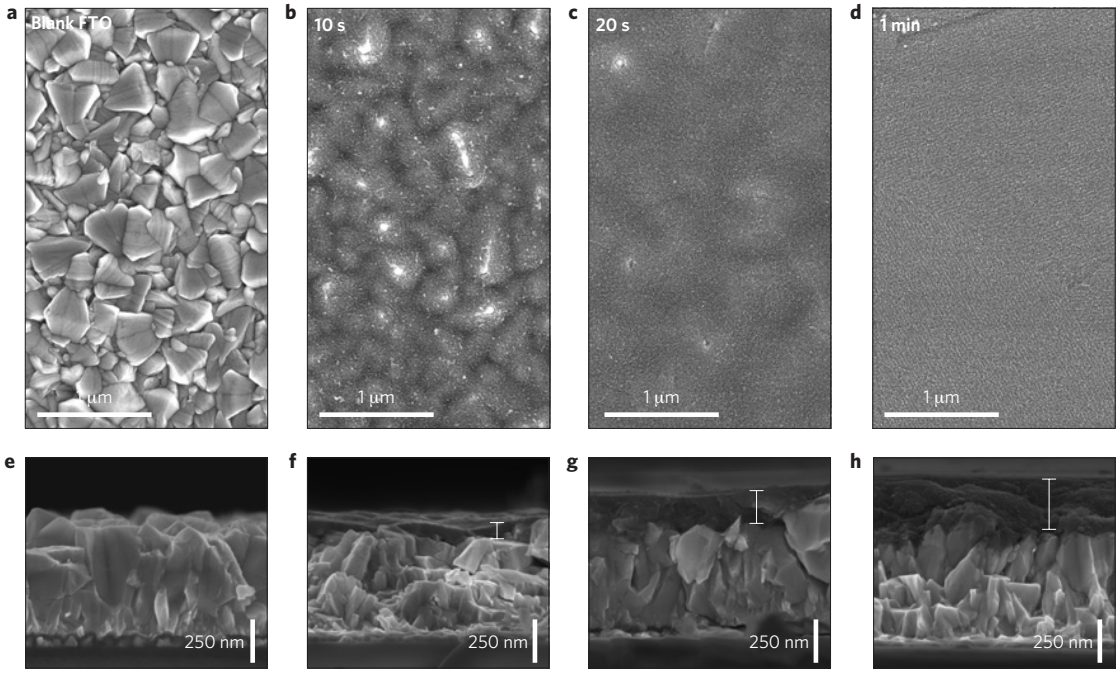

Figure 4 | SEM images of double-gyroid MoS_2 films. a–h, Top-down views perpendicular to the plane of the film (**a–d**) and cross-sectional views parallel to the plane of the film (**e–h**) for the clean FTO support (**a**,**e**) and the double-gyroid MoS_2 synthesized with a Mo deposition time of 10 s (**b**,**f**), 20 s (**c**,**g**) and 1 min (**d**,**h**).

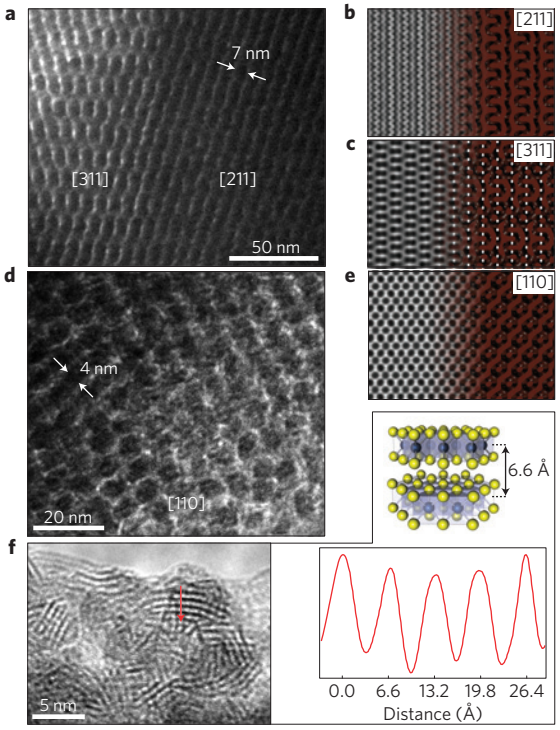

Figure 5 | TEM images of double-gyroid MoS_2. a, TEM image of double-gyroid MoS_2 where the [311] and [211] projections of the double-gyroid structure are observed. The pore-to-pore distance is ~7 nm as indicated by the white arrows. **b**,**c**, Models and corresponding simulated TEM images of the [211] and [311] projections, respectively. **d**, Zoom-in on the [211] and [110] projections showing the thickness of the double-gyroid MoS_2 networks. **e**, Model and corresponding simulated TEM image of the [110] projection **f**, TEM image where the S–Mo–S layers are resolved. The linescan indicated by the red arrow shows that the layer-to-layer spacing is 6.6 Å.

50 mV s^{-1}. The increased surface area for samples deposited with longer deposition times is evident when comparing the difference in capacitance current in the region of −0.1 V to +0.4 V versus the reversible hydrogen electrode (RHE). The sample deposited for 1 min shows the greatest capacitance whereas the sample deposited for 10 s shows the smallest capacitance. Comparing each sample's relative surface area (which also corresponds to relative thickness) as measured by means of capacitance (Supplementary Table S1) shows that the 20 s and 1 min samples have 2.4× and 4.3× the surface area of the 10 s sample, respectively. Each double-gyroid MoS_2 sample exhibited an onset for the HER at an overpotential of approximately 150–200 mV, consistent with previous reports for nanostructured MoS_2 (refs 15,18,20,38). Current density measurements (per geometric area) at −0.3 V versus the RHE reveal a clear trend that activity scales with thickness. This is expected owing to the higher electrochemically active surface areas for thicker samples. However, normalization of the data relative to each sample's surface area (Fig. 6b) shows that the increase in HER current does not scale linearly with surface area at higher current densities. In fact, the activity trend is reversed such that thinner films achieve higher turnover frequencies than thicker films.

Impedance spectroscopy (inset of Fig. 6b) revealed that the double-gyroid MoS_2 films themselves contributed significantly to the series resistance in addition to the substrate and solution resistances; consequently, this may contribute to the observed trends in turnover frequency. The high degree of resistance from the double-gyroid MoS_2 film is not surprising, given that MoS_2 possesses semiconducting as well as anisotropic charge transport properties[37]. As seen in the TEM image, the double-gyroid MoS_2 is composed of many grains, and there is no continuous pathway along any one layer to the substrate. This feature is desirable from the point of view of exposing an increased fraction of edge sites on the surface. However, the decreased conductivity limits the overall electrochemical performance for thicker films owing to diminished electron accessibility at the top of the film. This limitation can potentially be overcome by developing high-aspect-ratio conductive supports that are coated with a thin film of

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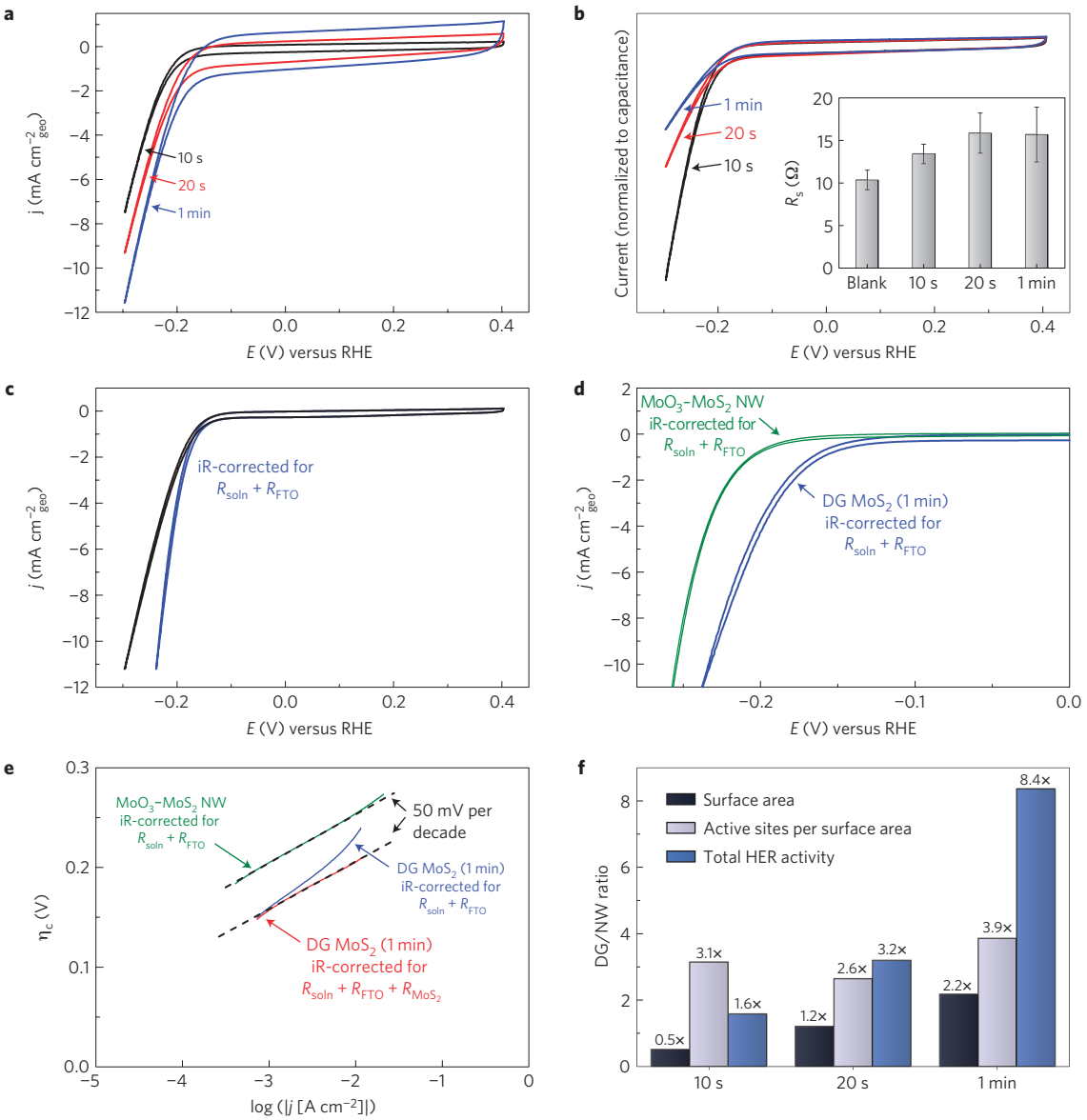

Figure 6 | HER activity of double-gyroid MoS_2. a, Cyclic voltammograms of 10 s, 20 s and 1 min samples at 50 mV s^{-1}. **b**, Comparison of the current in **a**, normalized to the capacitance of each sample. Inset: the measured R_s values from high-frequency impedance spectroscopy analysis. Error bars represent the 95% confidence interval (2 s.d.). **c** ,Cyclic voltammogram of 1 min sample at 5 mV s^{-1} with (blue) and without (black) correction for iR losses from the substrate and solution determined from the blank measurement in the inset of **b**. **d**, Comparison of the cyclic voltammogram of double-gyroid MoS_2 (1 min) versus core-shell MoO_3–MoS_2 nanowires (NW) at 5 mV s^{-1} (iR-corrected for resistances from the FTO substrate and the solution)[18]. **e**, Tafel plot of double-gyroid MoS_2 (1 min) versus core-shell MoO_3–MoS_2 nanowires, showing the 50 mV per decade slope once corrected for the resistance of the MoS_2. **f**, Ratios of surface area, density of active sites per surface area and total HER activity of the various double-gyroid MoS_2 films versus the nanowires.

double-gyroid MoS_2 to minimize charge transport distances while maintaining the total surface area, or by the addition of dopants to lower the resistivity of MoS_2 (ref. 39).

Nevertheless, the double-gyroid MoS_2 exhibits excellent activity for the HER. Figure 6c shows the activity of the 1 min sample when corrected for ohmic potential drop (iR) losses that arise only from the substrate and solution resistances. The iR-corrected activity for the double-gyroid MoS_2 surpasses the activity seen for high-aspect-ratio core–shell MoO_3–MoS_2 nanowires, as shown in Fig. 6d (ref. 18). However, two limitations distinguish the double-gyroid MoS_2: the slope of the current response is lower than the MoS_2 nanowires at higher current densities owing to iR losses from the more resistive double-gyroid structure and a visible build-up of large bubbles on the surface of the double-gyroid structure creates mass transport limitations that manifest as increased hysteresis between the cathodic and anodic sweeps, whereas the nanowires are better able to release bubbles from the surface (Supplementary Fig. S9). Mass transport limitations arising from gas bubble build-up could potentially be overcome by incorporating double-gyroid MoS_2 into a membrane electrode assembly[38] that does not use a liquid electrolyte or by the future development of appropriate surfactants to minimize bubble formation. On further correction of the j–V curve for iR losses through the MoS_2 itself, we see that all of the double-gyroid MoS_2 films exhibit a Tafel slope of approximately 50 mV per decade (Fig. 6e and Supplementary Fig. S6), consistent with previous studies of MoS_2 and suggesting a similar surface chemistry for the HER (refs 15,18). As the slope indicates that the surface chemistry is similar in nature between all of the samples, we employ a comparative Tafel analysis of the exchange current densities (Supplementary Fig. S6 and Table S1), yielding an overall

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enhancement of the HER activity relative to the nanowires of 1.6×, 3.2× and 8.4× for the 10 s, 20 s and 1 min double-gyroid MoS_2 films, respectively (Fig. 6f).

To decouple the enhanced activity of the double-gyroid MoS_2 films from contributions due to the surface area versus the density of active sites per surface area, we use impedance spectroscopy to assess the double-layer capacitance, which is proportional to the total electrochemically active surface area (details in Supplementary Information). When compared with a flat MoS_2 standard, the nanowires have a measured roughness factor (total sample surface area relative to geometric planar surface area) of 37. The roughness factors of the double-gyroid MoS_2 films were measured to be 19, 45 and 80 for the 10 s, 20 s and 1 min samples, respectively. These roughness factor values are in good agreement with theoretical models (Supplementary Fig. S7). This analysis reveals a 2–4× increase in the density of active sites per surface area for the double-gyroid films versus nanowires, enabling even the thinnest double-gyroid film (10 s, ∼50 nm) to outperform the nanowires by a factor of 1.6× despite having only half the surface area (Fig. 6f and Supplementary Table S1).

We have synthesized large-area contiguous thin films of a highly ordered mesoporous architecture of MoS_2 with a tunable film thickness that exhibit excellent catalytic activity for the HER. By controlling the MoS_2 morphology at the nanoscale, we engineered a change in the surface structure at the atomic scale and preferentially exposed more catalytically active edge sites, enabling improved performance. This particular material may find use in more traditional fields such as HDS in which MoS_2 edge sites have similarly been identified to be the active centres for catalysis[28,29,40–44]. Further improvements in the activity of this nanostructure could potentially be achieved on incorporation into a large macrostructured conductive support to minimize charge transport limitations and into a membrane electrode assembly to minimize mass transport limitations or by the addition of dopants to lower the resistivity of the MoS_2 film[39]. The addition of promoter atoms such as Co and Ni can further lead to enhancement in the catalytic activity for HDS and the HER (refs 20,45–50). Research efforts along these directions could potentially enable the double-gyroid MoS_2 to be able to compete against the best precious metal catalysts (such as Pt for HER) available today.

Methods

Synthesis of double-gyroid MoS_2. Double-gyroid MoS_2 was synthesized by electrodeposition of Mo into a silica template followed by sulphidization with H_2S. The silica double-gyroid template was prepared by using EO_{19}–PO_{43}–EO_{19} surfactant (Pluronic P84, BASF). The surfactant precursor solution was prepared by mixing 14.00 g of ethanol with 5.39 g of P84, which gives a Si/EO ratio of 1.20 (ref. 32). The precursor solutions were then sealed in a high-density polyethylene bottle and stirred gently at room temperature for 24 h. The silica solution was prepared by adding 12.86 g of ethanol, 6.35 g of a dilute aqueous 0.0177 M hydrochloric acid solution (pH 1.76) and 12.20 g TEOS to a high-density polyethylene bottle and stirred at room temperature for 20 min to initiate a prehydrolysis of the TEOS. The surfactant solution was then added to the silica solution to form the final coating solution. The coating solution was then left to age for 12 days at room temperature.

Films were dip-coated onto 1 cm × 3 cm FTO supports (sheet resistance: 8 Ω/□, Hartford Glass). FTO offers several advantageous properties for the purposes of this synthetic method. It remains intact when exposed to mild HF conditions for the short durations used during etching of the silica template and it can withstand exposure to the moderate sulphidization treatment used to synthesize the double-gyroid MoS_2 in this study. It is conductive, yet possesses a wide potential window of operation without exhibiting any significant electrochemical features and it remains stable in acidic electrolytes at moderate cathodic potentials. It is also a poor catalyst for the HER, enabling any activity from the MoS_2 catalyst to be clearly distinguished from the substrate. After coating, the films were dried at a controlled relative humidity of ∼50%, which enables evaporation-induced self-assembly of the surfactant into the double-gyroid structure. Finally the films were calcined in air at 400 °C for 4 h in a quartz tube furnace. Typical dip-coating speeds of 1 mm s^{-1} yielded calcined silica film thicknesses of approximately 300 nm by profilometry (Supplementary Fig. S1).

Molybdenum was electrodeposited into the double-gyroid silica template using a three-electrode electrochemical cell. A Ag|AgCl electrode was used as a reference electrode, and graphite was used as a counter electrode. The electrodeposition was carried out at a constant potential of −0.70 V versus Ag|AgCl and using different deposition times to control the thickness. The deposition solution consisted of an aqueous 50 mM peroxo-polymolybdate solution prepared by dissolving molybdenum metal (Aldrich) in 30% hydrogen peroxide (Fisher). These deposition conditions led to the formation of a mixed-valent molybdenum bronze ($Mo^{VI/V/IV}O_{3-x}(OH)_x$) within the double-gyroid silica template. After the electrodeposition, the samples were quickly removed from the deposition solution, rinsed with water, and dried to maintain film integrity.

The double-gyroid molybdenum bronze films were sulphidized with the silica template still present in a quartz tube furnace at 200 °C under a mixture of 10%/90% H_2S/H_2 gas (Matheson Trigas) flowing at approximately 60 s.c.c.m. (standard cubic centimetres per minute) for a period of 12 h using a 5 °C min^{-1} ramp rate. After sulphidization, the silica template was removed by etching with a 2 wt% HF solution for 30 min, leaving a bicontinuous inverse double-gyroid MoS_2 network. Afterwards, the network was briefly rinsed with water, dried, and re-sulphidized under the same flowing mixture of 10%/90% H_2S/H_2 gas at 200 °C for another 1 h.

Physical characterization. XRD was performed using a Philips PANalytical X'Pert Pro in parallel beam mode with Cu Kα radiation and 0.04 rad Soller slits. SEM was performed using an FEI Magellan 400 XHR microscope with a 5 kV accelerating voltage and 25 pA current. TEM was performed using an FEI Tecnai microscope with a 200 kV accelerating voltage. XPS was performed using a PHI VersaProbe XPS Microprobe with binding energies referenced to adventitious carbon at 284.6 eV. Raman spectroscopy was performed using a Horiba Jobin Yvon LabRAM Aramis system with a 532 nm excitation laser.

Electrochemical characterization. The electrochemical measurements were carried out in a three-electrode electrochemical compression cell (see Supplementary Fig. S10) using a Bio-Logic VSP potentiostat. The supported double-gyroid MoS_2 served as the working electrode with a graphite foil (Alfa) counter electrode and a $Hg|Hg_2SO_4$ (in saturated K_2SO_4) reference electrode (Hach). All cyclic voltammograms of the HER activity were conducted using a 0.5 M H_2SO_4 electrolyte under continuous sparging with 99.999% H_2 (Praxair). The reference electrode was calibrated to the reversible hydrogen potential using platinum mesh for both the working and counter electrodes in the same 0.5 M H_2SO_4 electrolyte sparged with 99.999% H_2 (Praxair). This calibration resulted in a shift of −0.704 V versus the RHE.

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Author contributions

J.K. and Z.C. carried out synthesis of the double-gyroid MoS_2 and performed XPS, XRD, SEM, Raman and electrochemical measurements. B.N.R. performed TEM imaging. J.K., Z.C. and T.F.J. conceived the study and wrote the manuscript.

Additional information

Supplementary information is available in the online version of the paper. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to T.F.J.

Competing financial interests

The authors declare no competing financial interests.

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